

Anthony Douglas.

Creative Software Developer — Real-Time Graphics & Audio

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SUMMARY

I'm a software developer focused on real-time graphics and audio. I write the GPU code behind the tools I create with — C++/CUDA and GLSL fluid solvers and shaders for TouchDesigner, a path tracer, and JamCanvas, a cross-platform audiovisual app with its own custom audio engine.

TECHNICAL SKILLS

Languages	C++17, GLSL, CUDA, TypeScript / JavaScript, Kotlin, Swift, Python
Graphics & GPU	OpenGL / GL ES, GLSL compute & fragment shaders, CUDA kernels, OptiX ray tracing, raymarching & SDFs, shader graphs, bloom / god-rays / post-processing
Audio DSP	Real-time synthesis, wavetable & modular synths, FFT / spectral analysis, lock-free audio threads, sequencers & samplers, JUCE (VST3 / CLAP)
Platforms & Tools	TouchDesigner Custom Operators (C++/CUDA POPs & TOPs), React, CMake, Android NDK / Oboe, iOS Core Audio, Git, Vercel

SELECTED PROJECTS

JamCanvas C++ · GLSL · Kotlin · Swift	Cross-platform AV app
Audiovisual jam app built on a shared C++ core with a custom lock-free audio engine (sequencer, samplers, wavetable & modular synths, FFT analyzer) and a GLSL render pipeline (raymarching, node-based shader graph, post-FX). Ships on SDL2/ImGui desktop, Android (NDK/Oboe), and iOS (Core Audio), with MIDI/OSC and NDI/Spout/Syphon output.	
jamcanvas-av.vercel.app github.com/dewpoint-av/JamCanvas	
FluidSPH GLSL · C++	Real-time fluid solver
GPU smoothed-particle-hydrodynamics fluid solver for TouchDesigner, with neighbor spatial hashing, running entirely in TouchDesigner's managed GPU context via GLSL compute — gives input geometry real fluid dynamics.	
github.com/dewpoint-av/FluidSPH	
AffectorPOP C++ · CUDA	TouchDesigner plugin
Native TouchDesigner custom operator (POP) that applies GPU particle forces — turbulence, vortex, attractor, drag — via the TouchDesigner C++ SDK and CUDA kernels. Part of a sellable suite of GPU operators.	
github.com/dewpoint-av/AffectorPOP	

EXPERIENCE

Independent Software Developer — Dewpoint AV	2024 – Present
Design, build, and sell GPU graphics and audio software: a suite of TouchDesigner custom operators (C++/CUDA/GLSL), an OptiX path tracer, and JamCanvas. Own the full stack from low-level GPU engines to React/TypeScript web and Patreon-based licensing.	
Audio / Video Engineer (A2 · V1) — Freelance	2023 – Present
Real-time AV systems, live visuals, and projection mapping for club- and festival-scale productions — a hands-on background in low-latency, real-time technical systems.	

EDUCATION

University of Tennessee Music Education	3 years
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